Nutr Cycl Agroecosyst (2025) 131:213–235
https://doi.org/10.1007/s10705-025-10418-4

ORIGINAL ARTICLE

Soil carbon accrual and yield response to biochar and compost in a four-year organic field study

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Received: 26 February 2025 / Accepted: 12 June 2025 / Published online: 4 July 2025
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Abstract Agricultural practices increasingly apply biochar through co-composting with organic fertilizers rather than via direct application. However, the agronomic assessment of these compost-biochar blends in organic farming, especially under repeated applications based on compost nitrogen (N) input, has not been comprehensively investigated. In a four-year organically managed field trial in southern Germany, the effects of annual biochar carbon (C) applications (0, 500, 1,000, 2,000 kg ha^{-1}) on crop yield, as well as changes in soil organic carbon (SOC) and total nitrogen (TN) stocks, were assessed. Biochar was applied either without additional fertilizers or co-composted with solid manure and grass-clover biomass (120 kg N ha^{-1}). Crops studied included spelt, grass-clover ley, winter wheat, and silage maize. Manure and grass-clover compost significantly increased spelt yields (e.g., control treatment: 3.21 t ha^{-1}; manure compost: 4.20 t ha^{-1}), winter wheat and silage maize yields. Co-composted biochar slightly reduced spelt yields in the first trial year (e.g., with manure compost, 4.20, 4.00, 3.98, 3.88 t ha^{-1} at biochar C levels of 0, 500, 1,000, and 2,000 kg ha^{-1}). No biochar effects were observed for grass-clover ley yield, but co-composted biochar marginally decreased winter wheat yields. In the fourth year of the trial, slight increase in maize yield was observed as a result of co-composting. Both compost and biochar applications led to increases in SOC stocks, with the applied biochar amounts matching the observed increases in SOC levels. The combination of biochar and compost resulted in SOC increases of up to + 10 t ha^{-1} after three trial years. These findings indicate potential long-term yield benefits resulting from enhanced soil properties.

Keywords Factorial field trial · Co-composted biochar · Organic agriculture · Crop yield · Soil organic carbon

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10705-025-10418-4.

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Introduction

Biochar is increasingly being utilized as a soil amendment. However, field trials under temperate climatic conditions in organically managed soils have been scarce. In organic farming, nitrogen (N) inputs are limited (no chemical N fertilizers, limited livestock densities and feed purchases). As a result, these

low-input systems often experience low N availability (Chmelíková et al. 2021), leading to a yield gap compared to conventional farming systems (Ponisio et al. 2015). Crop rotations in organic agriculture are based on the cultivation of legume crops, mainly grass-clover leys, for biological fixation of atmospheric N (Barbieri et al. 2023), which can result in N fixation rates of 100–500 kg ha^{-1} yr^{-1} (Braun et al. 2010; Pandey et al. 2017). Moreover, the use of high-quality compost is another key practice in organic farming (Erhart and Hartl 2010). Grass-clover biomass is traditionally used in livestock production, where farmyard manure is produced as a valuable fertilizer. In contrast, grass-clover is used as green manure or composted (grass-clover compost) in organic arable (stockless) farming, to provide N for non-legume crops. The optimization of the biological N fixation and the reduction of N losses during the storage and application of organic fertilizers are fundamental objectives within organic farming systems (Möller et al. 2008).

Biochar has been formally listed in the EU regulation on organic production and is conditionally permitted, provided it meets requirements regarding feedstock, chemical composition, and contaminant thresholds (European Union 2021). The potential benefits or drawbacks of using biochar in organic agriculture have been poorly discussed, as biochar may alter nutrient dynamics and consequently may lead to different yield responses in conventional and organic farming systems (Atkinson et al. 2010). In temperate climates, the use of biochar can contribute to nutrient immobilization processes and inhibited N turnover, potentially leading to yield reductions in non-legume crops (Borgen et al. 2012; Kayser et al. 2010; Nelissen et al. 2014). In contrast, biochar was shown to increase soil organic carbon (SOC), thereby improving soil fertility, yield potential, and contributing to climate protection, which are key goals of organic farming systems (Gattinger et al. 2012). Agronomic benefits may also arise from the fact that relevant amounts of nutrients are added along with biochar (depending on the application rate and the nutrient content) (Mia et al. 2014; van Zwieten et al. 2015). The mechanisms for potential yield increases also include improvements in soil chemical properties (liming effects) (Farhangi-Abriz et al. 2022). It must be noted that that these effects may be significantly more pronounced on highly weathered acidic or coarse-textured soils than on fertile temperate soils (Quilliam et al. 2013). Improving the conditions that favor legume N fixation could also be a potential merit. N immobilization processes stimulating biological N fixation of legumes have been demonstrated (Scheifele et al. 2017). However, the N transfer effects from legumes to non-legume crops in organic cropping systems with biochar amendment have been largely unexplored.

In agriculture, biochar is used as an additive in the composting processes, with a recommended addition of 10% by dry matter (DM) (Sanchez-Monedero et al. 2018). Experiments have shown that the addition of biochar accelerates the composting process (co-composting) and significantly reduces emissions of volatile N compounds and odors (Agyarko-Mintah et al. 2017; Sanchez-Monedero et al. 2018). Synergistic or antagonistic yield effects caused by co-composting have been described in the literature, based on different experimental approaches (Antonangelo et al. 2021; Seehausen et al. 2017). Internal on-farm analyses in Southern Germany indicate that only 50–200 kg of biochar ha^{-1} yr^{-1} are typically applied and accordingly, only a small amount of biochar is added during the composting process. The sole field application of pristine biochar is not common in farming practice (Schmidt et al. 2021a). Yet, biochar co-composting experiments have rarely been combined with field experiments for agronomic evaluation. Factorially designed agronomic trials with biochar, compost, and their mixture after co-composting have primarily been conducted in controlled environment pot experiments (Lebrun et al. 2023; Schulz et al. 2013; Seehausen et al. 2017). In field experiments, biochar and compost were not co-composted prior to application (Laird et al. 2017), or precise yield analysis was not conducted (Cooper et al. 2020). Compost provides essential nutrients, significantly influences crop yields, and has long-term effects on soil properties and processes (Hepperly et al. 2009). Therefore, a thorough assessment of its individual and synergistic effects with biochar amendment is necessary, alongside the inclusion of reference treatments with comparable nutrient inputs (Chaplot et al. 2025). Accordingly, the precise interpretation of the results regarding the effects of compost and biochar on soils and crops is currently limited (Seehausen et al. 2017). Results from a combined trial of biochar with organic and mineral fertilizers were presented by

Jaufmann et al. (2024) albeit in the context of conventionally managed intensive fertilization systems.

The temporal dynamics of biochar application must also be considered. Conceivably, the application of biochar may result in the long-term enhancement of soil properties by promoting the accumulation of SOC and total nitrogen (TN), altering soil C and N turnover, improving soil structure and aggregation, and ultimately influencing crop yield (Blanco-Canqui et al. 2020; DeCiucies et al. 2018; Weng et al. 2022). Collectively, the overall effect of biochar in combination with (co-composted) organic fertilizers on yield formation in temperate regions in organic farming is still largely unexplored, and it is unclear whether and after what application periods yield effects occur. The optimal combination of biochar and organic fertilizers is also unknown.

At the Roggenstein Experimental Station of the Technical University of Munich (TUM), a factorial field trial was established in 2020 with annual biochar application rates of 500–2,000 kg biochar C ha^{-1} yr^{-1}, both alone and co-composted with solid farmyard manure and grass-clover biomass. The management is based on the organic farming guidelines, and the trial is designed as a long-term field experiment, employing a crop rotation of spelt, grass-clover ley, winter wheat, and silage maize.

The results of the trials are expected to contribute to a better understanding and evaluation of the effects of biochar in organic farming systems:

- What yield impacts can organic farmers expect from biochar and biochar-compost applications? Will there be yield reductions or increases?
- How much of the applied biochar C is being detected in the soil? Do SOC and TN accumulate and are there interactions between compost and biochar C?
- What is the best mixing ratio of biochar and compost?

The individual and combined effects of biochar, manure compost, and grass-clover compost on biomass development, yield parameters, botanical grass-clover ley composition and yields, N contents and N use efficiency, and SOC and TN contents and stocks, as well as changes in SOC and TN contents and stocks during the experimental period were investigated.

Based on the current knowledge, the following hypotheses were tested:

1. Significant accrual of SOC is already detectable after three years of experimentation.
2. Manure and grass-clover composts enriched with biochar result in significant increases in non-legume crop yield compared to unfertilized treatments.
3. The combination of biochar with manure and grass-clover composts results in significant yield effects compared to compost-only amendments.
4. The application of pristine biochar (without fertilizers applied) leads to a significant decrease in non-legume crop yield due to N immobilization processes.

Materials and methods

Site conditions

The field trial was conducted at the Roggenstein Experimental Station, located 15 km west of Munich, Germany (48°11′25.8"N 11°20′09.7"E). The experimental site is located at an elevation of 510 m above sea level on diluvial colluvial soils derived from loamy sediments (Fluvic Cambisol) (Table 1). The soil type is silty clay loam, and no gravel (> 2 mm) is present.

During the 1991–2020 period, the mean annual temperature in Roggenstein was 8.9 °C, and the mean annual precipitation was 888 mm (Online Resource Table S22). The year 2021 was characterized by a cool spring and a wet summer with slightly above-average rainfall. In 2022, the mean annual temperature of 10.2 °C was 1.3 °C above the long-term average, reaching 10.2 °C, while the annual precipitation of 691.9 mm was well below normal, especially during the summer months. In 2023, there was average seasonal precipitation until mid-May, followed by a dry period with numerous hot days until shortly before the winter wheat harvest (DWD Climate Data Center 2025). The year 2024 proved to be optimal for maize growth, with above-average temperatures and abundant and well-distributed rainfall.

Table 1 Soil properties of the experimental plots measured at the baseline sampling. The values refer to the means across all experimental plots; the observed range of values (min–max) is given in parentheses

Parameter	Unit	0–30 cm	30–60 cm
Sand	%	33	n.d
Silt	%	55	n.d
Clay	%	12	n.d
Soil organic carbon (SOC)	g kg^{-1}	9.0 (8.3–10.0)	3.0 (2.2–4.0)
Total nitrogen (TN)	g kg^{-1}	0.8 (0.7–0.9)	0.3 (0.2–0.4)
C:N ratio		10.8 (10.1–12.2)	8.5 (7.0–9.5)
Soil pH ($CaCl_2$)		6.2 (5.8–6.8)	5.9 (5.6–6.5)
CAL-extractable phosphorus (P)	mg 100 g^{-1}	4.9 (3.4–6.1)	1.0 (0.6–1.5)
CAL-extractable potassium (K)	mg 100 g^{-1}	14.9 (11.7–18.0)	10.6 (8.8–12.1)
Bulk density	g cm^{-3}	1.44 (1.36–1.55)	1.61 (1.57–1.65)

Experimental design and layout

In previous decades, the field was used for conventional arable farming, without the application of organic fertilizers. The experimental area was converted to organic farming in 2020. The trial was conducted according to the internationally accepted guidelines for organic agriculture (European Union 2018; IFOAM 2018). The Roggenstein factorial field trial investigated the effects of four biochar application rates (0, 500, 1,000, and 2,000 kg biochar C ha^{-1} yr^{-1}), both in the absence of fertilization (U0, U500, U1000, U2000) and in combination with two types of compost: manure (M0, M500, M1000, M2000) and grass-clover compost (G0, G500, G1000, G2000), each supplying 120 kg total N ha^{-1} yr^{-1} (Table 2).

Two treatments combining 60 kg N ha^{-1} yr^{-1} of compost with 500 kg biochar C ha^{-1} yr^{-1} were implemented in the trial but were not included in the present study.

Biochar and organic fertilizers were applied annually prior to sowing: in late summer before grass-clover ley, in autumn before winter wheat and spelt, and in autumn to the catch crop preceding maize; the experimental procedures conducted are described in Online Resource Table S1. The crop rotation during the study period was spelt (*Triticum aestivum* subsp. *Spelta*, variety "Zollernspelz"), grass-clover ley, winter wheat (*Triticum aestivum*, variety "Wendelin"), and a catch crop followed by silage maize (*Zea mays*, variety "Pioneer 8329"). The grass-clover ley and catch crop (mainly clover species and sand oats) seeding mixtures are presented in Online Resource Tables S1 and S2. Each year, the field was plowed to

Table 2 Overview of treatments tested in the Roggenstein field trial. Twelve treatments were included. U0 represents the untreated control treatment. Treatments U500–U2000 receive only biochar, while M0 and G0 receive only compost (manure or grass-clover compost, respectively), without biochar amendment. M500–M2000 and G500–G2000 are co-composted biochar treatments. Factorial analysis was based on biochar application rate (0, 500, 1000, 2000 kg C ha^{-1} yr^{-1}) and fertilization type, which included the two levels unfertilized (U) and organic fertilization (M, G; 120 kg N ha^{-1} yr^{-1}). Post hoc tests were conducted within each fertilization group (U, M, G), comparing biochar levels (see Sect. "Statistical analysis")

	Unfertilized (+biochar)		Manure compost (+biochar)		Grass-clover compost (+biochar)	
Biochar C application rate (kg ha^{-1} yr^{-1})	Treatment	N input (kg ha^{-1} yr^{-1})	Treatment	N input (kg ha^{-1} yr^{-1})	Treatment	N input (kg ha^{-1} yr^{-1})
0	U0	0	M0	120	G0	120
500	U500	0	M500	120	G500	120
1,000	U1000	0	M1000	120	G1000	120
2,000	U2000	0	M2000	120	G2000	120

a soil depth of 25 cm prior to sowing. In this study, the effects on biomass formation, grain yield, and N uptake of spelt (2021); DM yield, N uptake and botanical composition of grass-clover ley (2022); biomass formation, grain yield, and N uptake of winter wheat (2023); DM yield, and N uptake of silage maize (2024), were investigated. Following a three-year trial period, soil samples were collected after the winter wheat harvest in 2023 to detect potential changes in SOC and TN stocks.

The fourfold replicated experimental treatments, each located in fixed experimental plots of 6 × 10 m, are arranged in a resolvable row-column design (Piepho et al. 2015). Each plot is divided into a sampling plot and a harvesting plot (3 × 10 m each). After harvest, cereal straw and grass-clover biomass were removed from the experimental plots. The schematic layout of the experimental plots (Figure S3) and the experimental design (Figure S4) are presented in the Online Resource.

Table 3 Properties of the biochar batches used

Parameter	Unit	2020	2021	2022	2023
Total organic carbon (TOC)	% DM	82.3	82.4	79.7	85.0
Specific surface area (BET)	$m^2\ g^{-1}$	238	237	276	n.a
Dry matter (DM)	%	71.5	68.0	61.4	68.1
pH		9.3	9.3	9.7	9.5
Total N (TN)	% DM	0.46	0.48	0.42	0.47
H/C_{org}		0.14	0.07	0.12	0.12
Particle size distribution					
> 10 mm	% (w/w)	2.4	3.0	0.2	0.1
8–10 mm	% (w/w)	3.4	2.3	0.4	0.1
6–8 mm	% (w/w)	4.3	3.0	0.6	1.0
4–6 mm	% (w/w)	4.2	8.0	3.3	1.9
2–4 mm	% (w/w)	3.5	6.4	6.4	3.7
1–2 mm	% (w/w)	6.8	6.9	21.2	6.6
< 1 mm	% (w/w)	75.4	70.3	68.0	86.7

Biochar production and characterization

The biochar used in the Roggenstein field trial was produced from wood residues in a combined slow pyrolysis and wood gasification system (SynCraft CW1000-300, Innsbruck, Austria). The maximum pyrolysis temperature was 850 °C. The pyrolysis plant was subjected to European Biochar Certification (EBC 2012–2023). The biochar used is characterized by a high C content, alkaline pH, and relatively fine particle size (Table 3).

Co-composting

The solid cattle manure (urine, dung, and straw) for the manure compost was obtained from a nearby organic dairy farm. At the beginning of the composting process, the moistened biochar was added to the manure piles and mixed using a tractor-mounted windrow compost turner. The grass-clover compost was produced from organically grown grass-clover biomass (90% w/w on a DM basis, grown at the organic experimental farm Viehhausen) and chopped miscanthus straw (10% w/w on a DM basis) as a bulking component. Freshly cut and trimmed grass-clover was piled with moistened biochar and miscanthus straw in compost heaps and mixed with a windrow compost turner. When necessary (due to a decrease in heap temperature caused by compaction), the compost heaps were turned (approximately 5–10 times, see Online Resource Figure S5) and covered with fleece to minimize weather effects and to ensure a homogeneous mixture. All composts were freshly made each year. The compost piles reached temperatures above 60 °C during the thermophilic phase (Online Resource Figure S5). The composting process was completed after approximately six to eight weeks (Kranert and Cord-Landwehr 2010). To determine the amount of biochar in the compost, mass balances were performed, assuming biochar C to be nearly stable (calculated biochar C decomposition and loss of 3%) (Wang et al. 2016). The exact biochar C amounts applied (Table S7) and the nutrient contents of the batches used (Table S8) are presented in the Online Resource. The composts and pristine biochar were applied manually before sowing to the experimental plots. Soil incorporation was performed immediately following field application to avoid wind drift. The tillage machinery used for soil incorporation and the depths of incorporation are presented in Table S1 (Online Resource).

Plant and soil sampling and laboratory analysis

A few days before harvest, 1.25 m^2 of spelt or winter wheat was cut manually from the sampling plot.

Yield parameters (stand density, kernels per ear, thousand grain weight) and straw yield were determined. The harvesting plot was harvested with a plot combine (Wintersteiger AG, Ried, Austria). The spelt grains were dehusked to determine spelt thousand kernel weight and grain yield. All yield data were normalized to 14% residual humidity. After grinding the grain and straw samples (< 0.5 mm), the N content was determined using an Elementar Vario MAX C/N Analyzer (Elementar Analysensysteme GmbH, Langenselbold, Germany). The grass-clover ley was harvested four times, with each harvest occurring at the onset of red clover flowering. Harvesting was done with a forage harvester, which determined the fresh matter yield. A sub-sample was analyzed for DM content. Biomass N content was determined using an Elemental Vario MAX C/N Analyzer. Additionally, 2 * 0.25 m^2 grass-clover biomass was cut by hand from selected treatments and sorted into clover (white and red clover), grasses, alfalfa, and weeds. Botanical composition was calculated from the DM of these species. The silage maize was harvested using a two-row chopper with an integrated weighing function, and biomass samples were used to determine the biomass DM and N content. Prior to the start of the experiment and following the winter wheat harvest, soil samples were collected at a fixed depth of 0–30 cm. Samples were collected immediately after cereal harvest (oats harvest, August 2020; winter wheat harvest, July 2023) without prior soil tillage from each experimental plot using a hand auger and, after drying, sieving (< 2 mm) and fine grinding, were analyzed for SOC and TN using a Skalar Primacs™ TOC/TN Analyzer (Skalar Analytical B.V., Breda, Netherlands). In addition, soil bulk density was determined using the core method to determine the total SOC and TN stocks (Blake and Hartge 2018). Both before the start of the field trial and after three experimental years, soil cores (100 cm^3 each, totaling 600 cm^3) were collected from the soil depth of 0–30 cm at 20 randomly selected undisturbed sites within the field trial to accurately map potential bulk density differences within the analyzed soil depth. Bulk density was calculated after drying for 24 h at 105 °C, using the mean value.

The SOC, TN, and inorganic N stocks were calculated using the following formula:

$$Stock_i\left[t/kg\ ha^{-1}\right] = Con_i * BD * LT \quad (1)$$

where i is SOC, TN, or inorganic N; Con is the SOC, TN, or inorganic N concentration (%); BD is the bulk density (g cm^{-3}, measured for SOC and TN stocks, estimated value of 1.5 for inorganic N stocks); and LT is the soil layer thickness sampled (30 cm each).

The soil inorganic N (NH_4-N + NO_3-N) stocks were determined at the beginning of each growing season (March), after harvest, and at the end of each growing season (November). Sampling was conducted using a tractor-mounted soil sampler, separately for fixed soil depths of 0–30 cm and 30–60 cm. The determination of NH_4-N and NO_3-N was conducted following the soil extraction with 0.1 M KCl solution, employing spectrophotometry (in accordance with the standards set forth in DIN ISO 14256).

N use efficiency was calculated by subtracting the biomass N uptake measured in the unfertilized control treatment from the biomass N uptake of each fertilized treatment, and dividing the result by the compost N input.

$$N\ use\ efficiency = \left(\frac{N\ uptake\ (fertilized\ treatment) - N\ uptake\ (unfertilized\ control)}{N\ input}\right) * 100 \quad (2)$$

Spectral measurements

Spectral measurements were conducted in spelt and winter wheat using a handheld multispectral sensor (tec5, Oberursel, Germany) and the vegetation index REIP (Red-Edge Inflection Point) was subsequently calculated (Guyot et al. 1988). The REIP index is positively correlated with biomass yield, N content, and N uptake (Rischbeck et al. 2016), and allows for non-destructive measurements during the growing season at relevant phenological growth stages. The calculation of the REIP is based on the following formula:

$$REIP = 700 + 40 * \left(\frac{\left(\frac{R670+R780}{2}\right) - R700}{R740 - R700}\right) \quad (3)$$

The measurements were made up of eight to ten individual data points per plot. The measurements were conducted weekly from the beginning

of tillering until the end of flowering of spelt and winter wheat. A tractor-mounted platform was used to carry out measurements at the flowering stage in silage maize.

Statistical analysis

Statistical analysis and figure creation was conducted using the RStudio software (R Core Team 2022). Data analysis was performed using mixed models (Gałecki and Burzykowski 2013). The following packages were used for data analysis: "dplyr", "lme4", and "lmertest". For crop yield, SOC, and TN analysis, the experimental factors "biochar" and "organic fertilization" (including manure and grass-clover compost) and their interaction were set as fixed factors (Table 2). In contrast, the replications within the experimental area (experimental row (1–4), experimental column (A-D), and experimental block (a-d), see experimental design (see Online Resource Figure S4)) were designated as random factors, with the objective of eliminating the influence of soil heterogeneity. The data obtained from the two treatments that received 60 kg compost N ha^{-1} yr^{-1} were excluded from the data set in this study. The botanical composition of the grass-clover ley was analyzed using data from the eight treatments studied (unfertilized + biochar, manure compost + biochar). In addition to the factors "biochar" and "organic fertilization", the species group (clover, alfalfa, grasses, weeds) was included as an additional factor in the model. Each cut was evaluated separately. Non-significant interactions of the models were removed. The Akaike Information Criterion (AIC) was calculated to determine the model with the best fit. The model with the lowest AIC was used to calculate p-values (Gałecki and Burzykowski 2013). A graphical residual analysis was then performed. The significance of the factors and interactions is characterized as follows: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, n.s. not significant. Furthermore, pairwise comparisons between treatments within a fertilization level (unfertilized + biochar, manure compost + with biochar, grass-clover compost + biochar) were performed using the "emmeans" package (Lenth 2024). Significant differences between treatments ($p < 0.05$) are indicated by different lowercase letters.

Results

Changes in SOC and TN contents and stocks

At the outset of the trial, the SOC content averaged 9.0 g kg^{-1} across all treatments. After three trial years, an increase in SOC content up to 11.3 g kg^{-1} was observed in the biochar- and compost-amended treatments. SOC content increased with increasing application rates of biochar (Table 4). Considering the soil bulk density, after three trial years, the SOC stock in the unfertilized control treatment increased by 0.99 t ha^{-1}. Organic fertilization with manure and grass-clover composts (without biochar admixture) significantly ($p < 0.01$) increased SOC stocks to 2.92 and 4.61 t ha^{-1}, respectively, which is a further increase of 1.93 and 3.62 t ha^{-1}.

Depending on the biochar C rate applied, pristine or co-composted biochar significantly increased SOC stocks: Adding 1.5 t ha^{-1} of biochar C (without composts) in the three trial years further increased SOC stock by 0.84 t ha^{-1} to 1.83 t ha^{-1} compared to the control treatment. Adding 3.0 t ha^{-1} of biochar C increased SOC stock by 3.98 (+2.99) t ha^{-1}, and adding 6.0 t ha^{-1} increased SOC stock by 8.26 (+7.27) t ha^{-1} (Online Resource Table S17). A similar pattern was observed with biochar co-composting: At a biochar C application rate with manure compost of 1.28 t ha^{-1} (targeted rate: 1.5 t ha^{-1}), SOC stock increased by 4.30 t ha^{-1} to 7.22 t ha^{-1} compared to treatment receiving manure compost-only. With twice the amount of biochar applied (2.56 t ha^{-1} after three years, targeted rate: 3.0 t ha^{-1}), the SOC stock increased by 3.88 t ha^{-1} to 6.80 t ha^{-1}. The highest biochar C application rate (5.84 t ha^{-1} after three years, targeted rate: 6.0 t ha^{-1}) resulted in an increase in SOC stock of 9.79 (+6.87) t ha^{-1}.

With the grass-clover compost amendment, the SOC stock initially decreased due to the application of 1.41 t ha^{-1} biochar C (by 1.39 t ha^{-1} to 3.22 t ha^{-1} SOC stock compared to the unamended grass-clover compost (SOC stock increase of 4.61 t ha^{-1})). However, at higher biochar C rates (2.82 and 4.79 t ha^{-1}, at targeted rates of 3.0 and 6.0 t ha^{-1}), the SOC stocks

Table 4 SOC and TN contents at the outset of the trial (Baseline sampling, 2020) and following three years of experimentation (fixed soil depth, 0–30 cm). Based on these values, the soil C:N ratio was calculated. All data is expressed as the treatment mean ± standard deviation, post-hoc tests have been conducted for soil C:N ratio

Parameter	Cumulated biochar C input (2020–2023)	Cumulated compost N input (2020–2023)	SOC content (Baseline sampling, 2020)	SOC content (after three trial years, 2023)	TN content (Baseline sampling, 2020)	TN content (after three trial years, 2023)	Soil C:N ratio 2023
Unit	t ha^{-1}	t ha^{-1}	g kg^{-1}	g kg^{-1}	g kg^{-1}	g kg^{-1}	
Unfertilized (+ biochar)	0		8.9 ± 0.6	9.1 ± 0.5	0.79 ± 0.05	0.78 ± 0.05	11.7 ± 0.5ab
	1.5		8.8 ± 0.4	9.3 ± 0.5	0.76 ± 0.04	0.82 ± 0.06	11.4 ± 0.5a
	3.0		8.7 ± 0.4	9.7 ± 0.5	0.78 ± 0.03	0.83 ± 0.03	11.7 ± 0.3ab
	6.0		8.8 ± 0.7	10.8 ± 0.6	0.78 ± 0.05	0.86 ± 0.05	12.5 ± 0.2b
Manure compost (+ biochar)	0	0.36	9.1 ± 0.4	9.9 ± 0.1	0.79 ± 0.02	0.88 ± 0.01	11.3 ± 0.1a
	1.28	0.36	9.0 ± 0.5	10.8 ± 0.4	0.79 ± 0.03	0.90 ± 0.04	12.0 ± 0.3b
	2.56	0.36	8.9 ± 0.4	10.6 ± 0.3	0.75 ± 0.10	0.90 ± 0.05	11.9 ± 0.6ab
	5.84	0.36	8.9 ± 0.4	11.3 ± 0.5	0.78 ± 0.04	0.90 ± 0.03	12.6 ± 0.6b
Grass-clover compost (+ biochar)	0	0.36	9.1 ± 0.5	10.5 ± 0.6	0.80 ± 0.05	0.91 ± 0.07	11.5 ± 0.6a
	1.41	0.36	9.2 ± 0.5	10.1 ± 0.3	0.80 ± 0.04	0.87 ± 0.06	11.5 ± 0.5ab
	2.82	0.36	9.1 ± 0.4	10.7 ± 0.3	0.80 ± 0.05	0.91 ± 0.02	11.9 ± 0.4ab
	4.79	0.36	9.1 ± 0.5	11.3 ± 0.3	0.80 ± 0.05	0.91 ± 0.04	12.4 ± 0.6b
Factors	Biochar						***
	Org. Fertilizer						n.s
	Biochar x Org. Fertilizer						n.s

increased from 1.83 t ha^{-1} to 6.44 t ha^{-1} and by 4.58 t ha^{-1} to 9.19 t ha^{-1}, respectively.

The deviation of SOC stock changes within each treatment is important to note, however (Table 4). The analyses revealed no evidence of an interaction effect between the application of biochar and compost. The baseline sampling showed an average soil TN content of 0.79 g kg^{-1}. After three consecutive trial years, soil TN content increased significantly to 0.88 g kg^{-1} and 0.91 g kg^{-1} due to the manure and grass-clover compost amendments, respectively. The measured TN stock slightly decreased by 0.08 t ha^{-1} in the unamended control treatment after three trial years. Compost application significantly ($p < 0.001$) increased the soil TN stocks. Interestingly, a significant increase in TN stocks was observed with biochar amendments applied alone and in combination ($p < 0.01$) with the manure compost (Fig. 1, Table S17). As a result of biochar application, soil C:N ratios increased significantly, whether applied alone or in combination with composts (Table S17).

Spelt, 2021

Grain yield, N uptake and yield parameters

In the first trial year, the spelt grain yield in the unfertilized treatment was 3.21 t ha^{-1}. The application of manure compost significantly increased the spelt grain yield to 4.20 t ha^{-1}, and the application of grass-clover compost to 4.17 t ha^{-1} ($p < 0.001$). As biochar application rates increased, spelt grain yield showed a slight but statistically significant decline, both applied alone and in combination with composts (with manure compost: 4.20, 4.00, 3.98, 3.88 t ha^{-1} at biochar C levels of 0, 500, 1,000, and 2,000 kg ha^{-1}). Similarly, with grass-clover compost, spelt grain yield initially decreased with increasing biochar application rates, then increased again at the highest biochar level (Table 5).

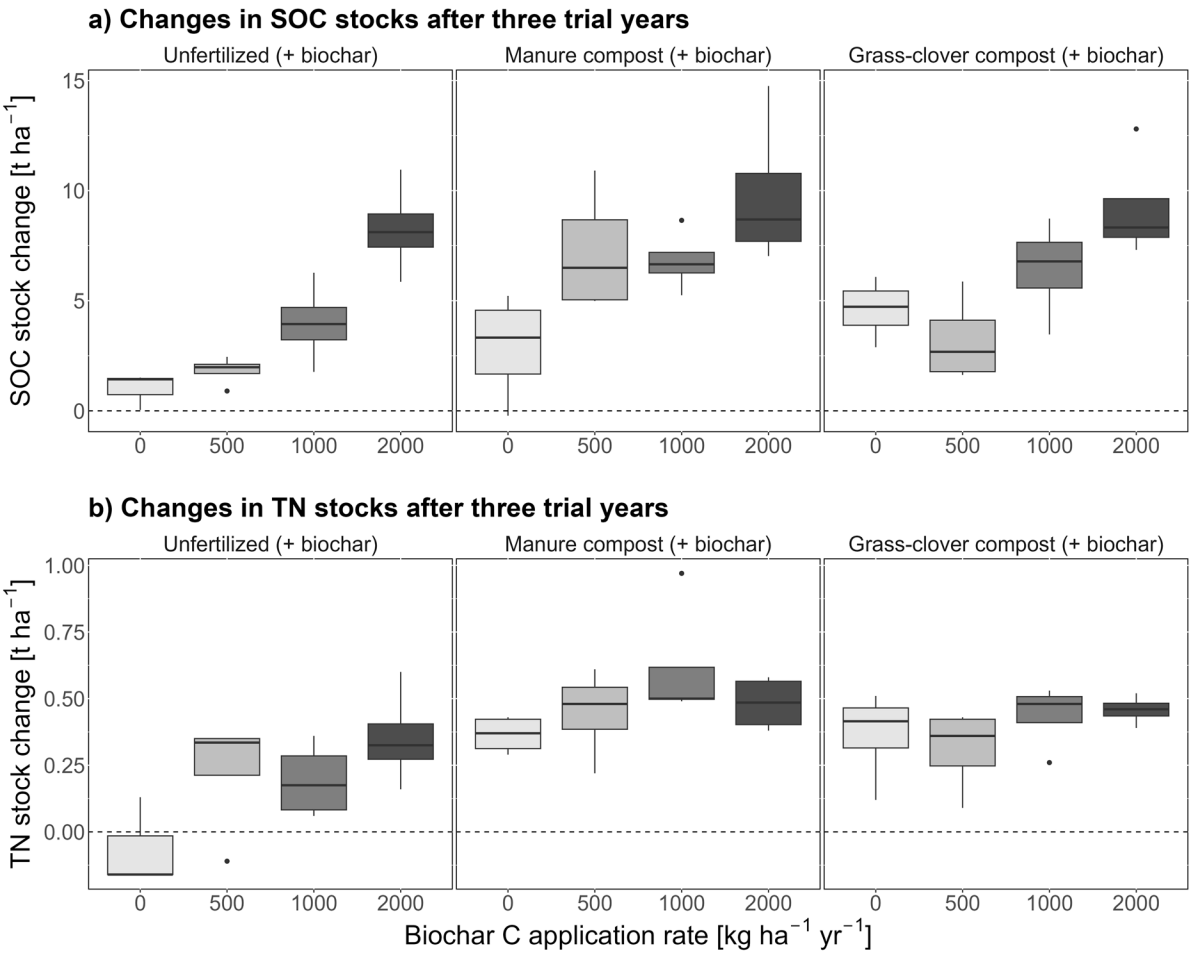


Fig. 1 Changes in SOC (**a**) and TN stocks (**b**) following three consecutive trial years with consideration of the bulk density (averaged over all treatments: 1.454 g cm^{-3} in the baseline sampling (2020), 1.442 g cm^{-3} for the 2023 sampling)

Spelt N uptake in the unamended control was 58.0 kg ha^{-1}, and spelt N uptake receiving manure compost without biochar was 74.9 kg ha^{-1}. However, N uptake decreased with increasing biochar C content in the manure compost, reaching 72.9, 67.7, and 66.8 kg ha^{-1} (at biochar C rates of 500, 1,000, and 2,000 kg ha^{-1}), respectively. A similar pattern was observed with the grass-clover compost, with N uptake decreasing at a biochar C application rate of 500 kg ha^{-1} but increasing at biochar C levels of 1,000 and 2,000 kg ha^{-1}. No interaction effect for biochar and organic fertilization was observed. Applying pristine biochar also exhibited a slight but consistent and significant reduction in N uptake (53.2, 51.6, and 51.1 kg ha^{-1} at biochar C levels of 500, 1000, and 2,000 kg ha^{-1}, respectively, as shown in Table 5). Manure compost exhibited an N use efficiency of 14.1%, while that of the grass-clover compost was 10.2%. As the biochar content in the manure compost increased, the N use efficiency gradually decreased (12.4, 8.1, 7.4%). The N use efficiency of grass-clover compost decreased at the 500 kg ha^{-1} biochar C rate (8.0%), then increased (not significantly) at higher biochar C quantities to 9.5, and 10.7%, respectively.

The application of organic fertilizers, including manure and grass-clover compost, significantly increased yield parameters such as stand density, number of grains per spike, and thousand grain weight ($p < 0.001$). However, biochar application did not significantly affect these parameters, except for reduced grains per ear due to 1,000 kg ha^{-1} biochar C co-applied with grass-clover compost.

Table 5 Spelt grain yield, N uptake, N use efficiency, and yield structure parameters (2021, mean + standard deviation). Significant differences ($p < 0.05$) between the treatments are indicated by different letters

Parameter	Biochar C application rate	Compost N application rate	Grain yield (86% DM)	N uptake (grain + straw)	N use efficiency	Stand density	Grains per ear	Thousand grain weight (86% DM)
Unit	kg ha^{-1} yr^{-1}	kg ha^{-1} yr^{-1}	t ha^{-1}	kg ha^{-1}	%	ears m^{-2}		g
Unfertilized (+biochar)	0		3.21 ± 0.1	58.0 ± 3.2a		363.6 ± 13.8	16.7 ± 2.3	34.7 ± 1.1
	500		3.14 ± 0.2	53.1 ± 5.4ab		352.5 ± 23.4	18.3 ± 0.7	35.1 ± 1.0
	1000		3.06 ± 0.2	51.6 ± 3.9ab		375.5 ± 21.9	15.8 ± 1.8	35.0 ± 0.8
	2000		3.05 ± 0.2	51.1 ± 4.7b		341.7 ± 45.2	19.1 ± 2.9	34.4 ± 0.4
Manure compost (+biochar)	0	120	4.20 ± 0.1	74.9 ± 3.9	14.1 ± 3.3	426.9 ± 5.1	21.6 ± 1.5	37.1 ± 0.9
	500	120	4.00 ± 0.2	72.9 ± 5.9	12.4 ± 4.9	386.7 ± 39.0	21.2 ± 3.8	36.4 ± 1.2
	1000	120	3.98 ± 0.3	67.7 ± 6.8	8.1 ± 5.7	388.5 ± 51.7	21.9 ± 0.9	36.3 ± 0.5
	2000	120	3.88 ± 0.3	66.8 ± 8.4	7.4 ± 7.0	371.8 ± 10.3	22.1 ± 2.3	35.8 ± 0.9
Grass-clover compost (+biochar)	0	120	4.17 ± 0.3	70.3 ± 7.0	10.2 ± 5.9	380.3 ± 22.5	22.6 ± 2.9a	35.8 ± 0.4
	500	120	3.98 ± 0.4	67.6 ± 4.8	8.0 ± 4.0	406.7 ± 25.3	21.6 ± 2.5ab	35.8 ± 0.7
	1000	120	3.83 ± 0.3	69.4 ± 6.5	9.5 ± 5.4	393.8 ± 29.2	19.2 ± 3.0b	36.0 ± 1.0
	2000	120	4.05 ± 0.3	70.8 ± 5.9	10.7 ± 4.9	393.7 ± 22.8	21.2 ± 1.4ab	35.8 ± 1.0
Factors	Biochar		*	*	n.s	n.s	n.s	n.s
	Org. Fertilizer		***	***		***	***	***
	Biochar x Org. Fertilizer		n.s	n.s		n.s	n.s	n.s

Vegetation index REIP Significant effects of biochar and compost fertilization on the vegetation index REIP were observed in the early vegetation stages (see Online Resource Table S9). REIP increased with the application of manure and grass-clover compost, while biochar addition, both applied alone and co-composted, slightly but consistently reduced REIP. The results of the spectral measurements correlated closely with the measured yield data (up to $r = 0.92$, Fig. 2, Online Resource Table S10). The correlation coefficient initially increased until BBCH 55 (phenological stage) and then declined. However, the REIP-increasing effect of compost and the REIP-reducing effect of biochar were evident throughout the main growing season.

Grass-clover ley, 2022

Dry matter yield, N uptake, and botanical composition

The DM yields of the four cuts of the grass-clover ley averaged across all treatments were 3.6, 5.3, 3.2, and 2.4 t ha^{-1}, with a total DM yield of 14.6 t ha^{-1}. A significant effect ($p < 0.01$) of organic fertilization was observed in the first and fourth cut, as manure compost application slightly increased DM yield. Consequently, a slight effect of organic fertilization was observed when considering the total DM yield. Pristine or co-composted biochar had no significant effect on grass-clover ley yield (Online Resource Table S11). N uptake of the aboveground biomass was 482.9 kg ha^{-1} including all four cuts, with the highest N uptake occurring in the second growing period (194.5 kg ha^{-1} on average). Biochar, whether applied alone or co-composted, significantly increased the N uptake for the second cut, and significant increases were also observed for the fourth cut. However, neither biochar application nor organic fertilization significantly affected total N uptake. Manure compost application shifted the botanical composition in favor of non-leguminous species. Applying biochar, either alone or in combination with manure through co-composting, did not significantly alter the botanical species composition.

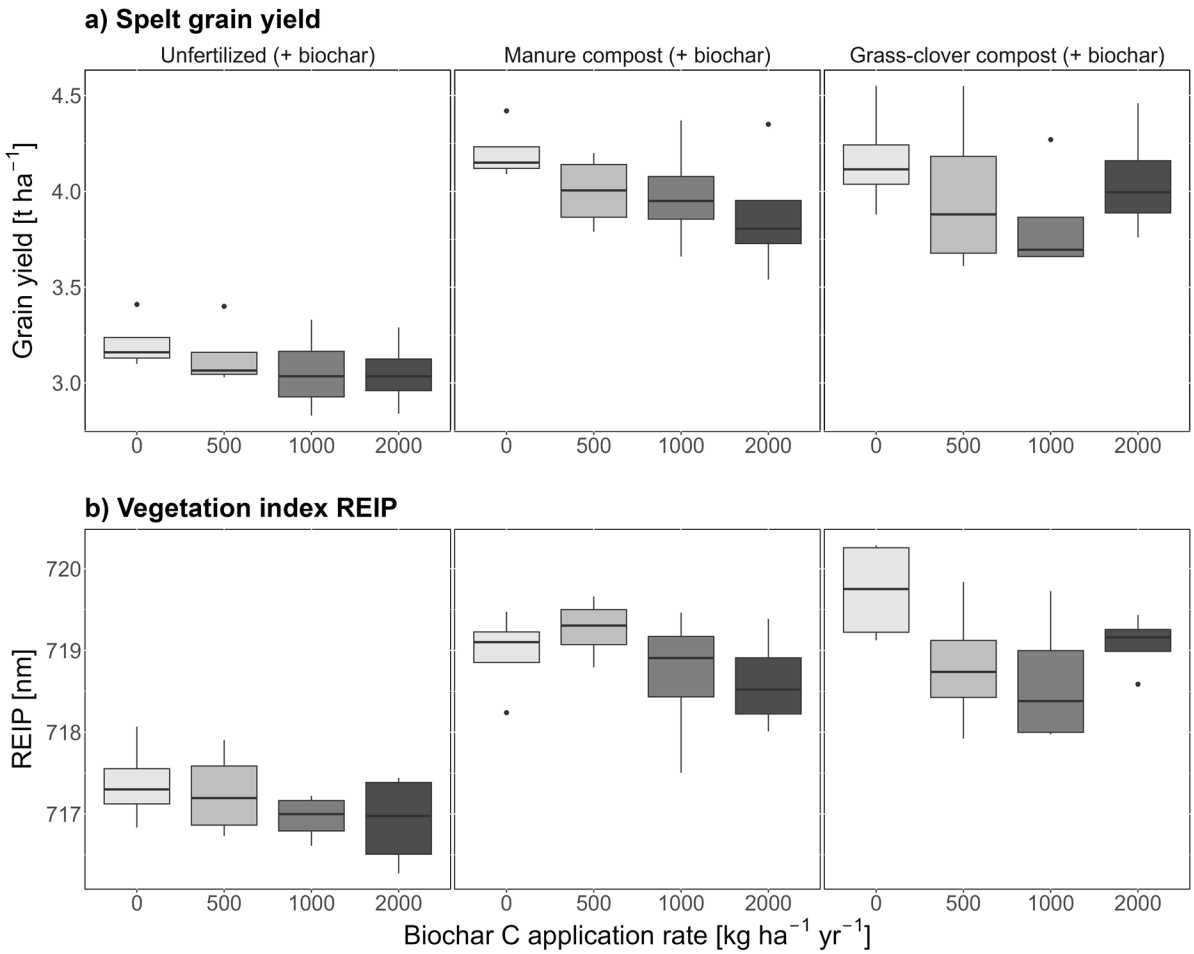


Fig. 2 Evaluation of the vegetation index REIP (**b**) measured on May 28, 2021, and its relation to spelt grain yield (**a**)

Winter wheat, 2023

Grain yield, N uptake, and yield parameters

In the unfertilized treatment, the winter wheat grain yield was 4.97 t ha^{-1}. With the application of manure and grass-clover compost, the grain yield increased significantly ($p < 0.001$) to 5.82 and 6.13 t ha^{-1}, respectively. No significant biochar effect was observed either when applied alone or in combination with compost. However, the combination of biochar with grass-clover compost resulted in a slight and consistent, though non-significant, decrease in grain yield (6.13, 5.92, 5.85, 5.82 t ha^{-1} at biochar C application rates of 0, 500, 1,000, 2,000 kg ha^{-1}). For manure compost, the lowest grain yield was observed with manure compost-only (5.82 t ha^{-1}). The addition of a small quantity of biochar C (500 kg ha^{-1}) resulted in an increase in grain yield to 6.35 t ha^{-1}. However, as the biochar rate was increased, the grain yield decreased, although not significantly (6.21 and 6.14 t ha^{-1} at biochar C application rates of 1,000 and 2,000 kg ha^{-1}) (Online Resource Table S13). The application of organic fertilizers, including manure and grass-clover compost, resulted in a significant increase in N uptake ($p < 0.001$). However, the addition of biochar did not significantly influence compost N use efficiency, which remained approximately 20%. The increase in winter wheat grain yield observed in the compost treatments was attributed to significantly increased stand density and significantly higher thousand grain weight ($p < 0.001$), while the number of grains per ear was

not significantly affected by any of the factors studied (Table S13).

Vegetation index REIP

The vegetation index REIP was not significantly affected by biochar amendment in the absence of compost fertilization at any measurement. However, the application of grass-clover compost and manure compost resulted in a highly significant increase in the REIP values at each measurement ($p < 0.001$). A significant interaction effect of biochar and organic fertilization was observed in most measurements, characterized by a slight but consistent reduction of the REIP due to co-composted biochar (e.g., REIP, May 13; 725.2, 724.7, 724.6, 724.3 nm at 0, 500, 1,000, 2,000 kg biochar C ha^{-1} in grass-clover compost) (Online Resource Figure S14, Table S15). The correlation between REIP and wheat yield was up to r = 0.94 (Online Resource Table S16).

Silage maize, 2024

In the untreated reference treatment, silage maize yielded 14.5 t ha^{-1}, with an N uptake of 125.5 kg ha^{-1}. Although slight but significant variations in crop yield occurred with increasing biochar C application without compost, no distinct pattern was evident (Table 6). In contrast to previous years, all biochar-compost treatments resulted in increased DM yield and N uptake in maize compared to compost-only treatments, with a significant interaction between the two factors. In the manure compost treatments, the highest DM yield (20.0 t ha^{-1}) and N uptake (182.5 kg ha^{-1}) were observed at 500 kg ha^{-1} biochar C, whereas higher biochar application rates led to a non-significant yield decline (Fig. 3). In contrast, maize yield and N uptake increased with higher biochar application rates in the grass-clover compost treatment, with yield increases approaching statistical significance ($p < 0.1$). N use efficiency was 27.2 and 37.3% for manure and grass-clover compost, respectively. There was a corresponding non-significant increase in N use efficiency in line with the amount

Table 6 Silage maize biomass DM yield, N uptake, N use efficiency, and REIP (2024, mean + standard deviation). Significant differences ($p < 0.05$) between the treatments are indicated by different letters

Parameter	Biochar C application rate	Compost N application rate	Biomass DM yield	Biomass N uptake	N use efficiency	REIP
Unit	kg ha^{-1} yr^{-1}	kg ha^{-1} yr^{-1}	t ha^{-1}	kg ha^{-1}	%	nm
Unfertilized (+ biochar)	0		14.5 ± 0.6ab	125.5 ± 19.0		723.0 ± 0.6
	500		14.0 ± 1.3a	113.6 ± 8.5		723.4 ± 0.3
	1,000		15.3 ± 1.0b	125.8 ± 13.4		723.2 ± 0.5
	2,000		14.3 ± 1.0ab	115.1 ± 10.7		723.5 ± 0.5
Manure compost (+ biochar)	0	120	17.0 ± 2.4	158.1 ± 18.7	27.2 ± 15.6	725.2 ± 1.0
	500	120	20.0 ± 1.1	182.5 ± 11.8	47.5 ± 9.8	725.6 ± 0.5
	1,000	120	18.7 ± 2.2	168.2 ± 33.4	35.6 ± 27.8	725.4 ± 0.7
	2,000	120	18.0 ± 2.1	152.0 ± 14.6	22.1 ± 12.1	725.3 ± 0.9
Grass-clover compost (+ biochar)	0	120	16.5 ± 2.9	170.2 ± 19.6	37.3 ± 16.3	724.9 ± 0.7
	500	120	16.6 ± 1.4	159.3 ± 17.9	28.2 ± 14.9	724.9 ± 0.2
	1,000	120	18.6 ± 2.2	180.9 ± 27.8	46.2 ± 23.1	725.5 ± 0.8
	2,000	120	19.5 ± 0.7	177.0 ± 3.4	42.9 ± 2.9	725.8 ± 0.6
Factors	Biochar		*	n.s	n.s	n.s
	Org. Fertilizer		***	***		***
	Biochar x Org. Fertilizer		*	n.s		n.s

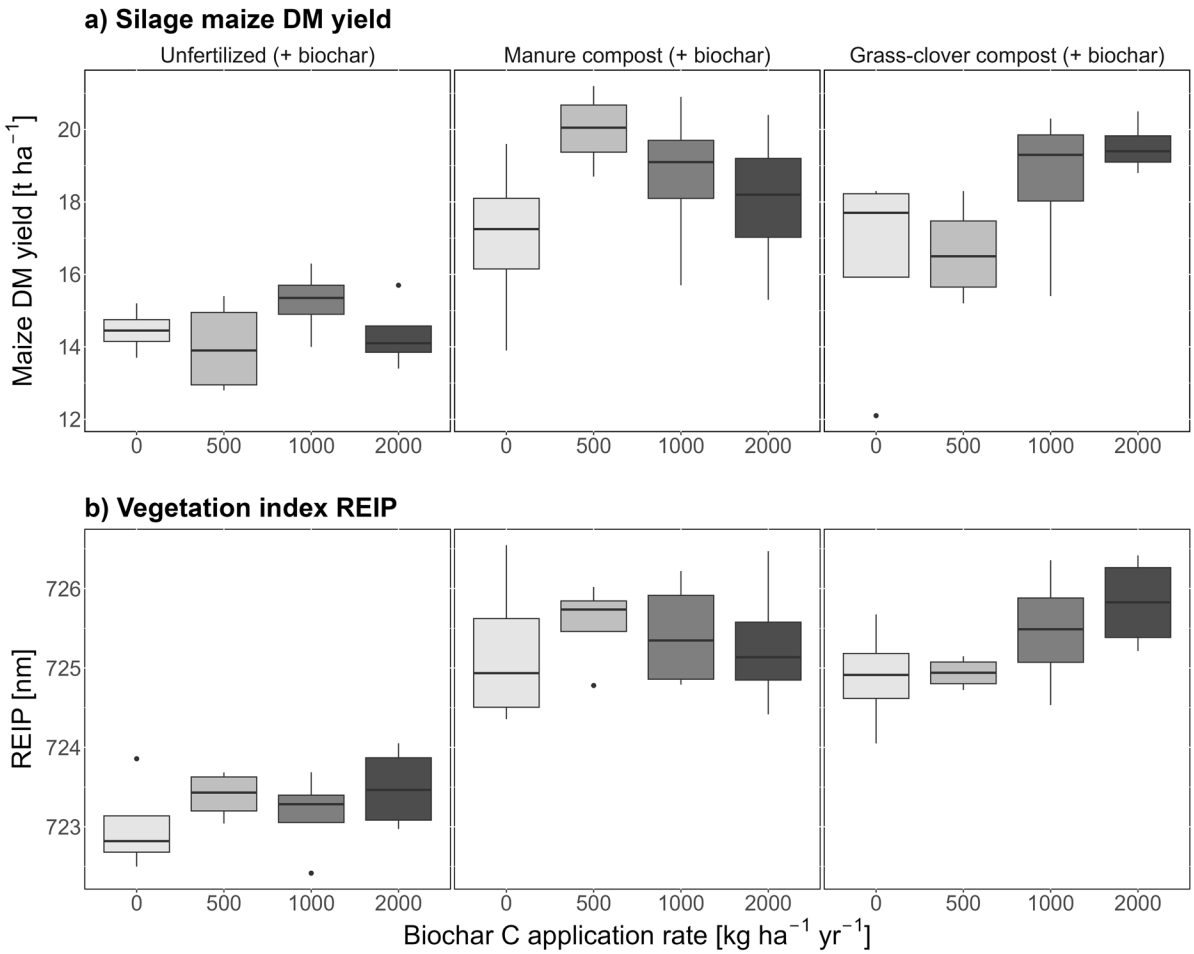


Fig. 3 Evaluation of the vegetation index REIP (**b**) measured on July 25, 2024, and its relation to silage maize DM yield (**a**)

of biochar C added. The REIP vegetation index indicates the beneficial effects of compost application, as well as the non-significant positive trends observed with biochar co-composting. Unlike the yield data, no significant effects were detected for the application of pristine biochar (Fig. 3). Accordingly, the correlation between REIP and yield data in maize was lower ($r = 0.83$) than for spelt or winter wheat.

Discussion

Discussion of methods

In previous experiments, the effects of co-composting biochar were studied without analyzing the agronomic traits under field conditions (Busch and Glaser 2015). In this experiment, the effects of manure and grass-clover composts with different combinations of biochar (co-composted) were examined. To distinguish between the effects of biochar and those of the compost (Seehausen et al. 2017), a factorial experimental design including valid reference treatments was developed (Chaplot et al. 2025). This approach allows the effects of co-composted biochar to be examined, as well as the individual effects of biochar, manure compost, and grass-clover compost and to determine the optimal, practice-relevant application rates of co-composted biochar. The experiment is conducted in accordance with the guidelines of organic farming; the conversion to organic farming started with the establishment of the experiment. Consequently, residual effects may occur in the transition phase.

The soils at the Roggenstein experimental station have been used for conventional arable farming for decades (mineral fertilization only, no organic fertilization, no legumes), resulting in low SOC stocks, and low N mineralization potential (Mittermayer et al. 2024). The initial SOC and TN contents can have a significant influence on the SOC and TN dynamics, as numerous long-term field experiments show (Stewart et al. 2008). At low initial SOC and TN contents (as in Roggenstein), organic fertilization is expected to result in strong SOC and TN accumulation. Changes in SOC stocks resulting from biochar amendment are primarily influenced by the amount of biochar applied, owing to its recalcitrant nature and minimal microbial decomposition. In contrast, noticeable changes in SOC and TN stocks from compost application under realistic farm-level conditions are typically detectable only after several years or decades of use (Kätterer et al. 2014; Körschens et al. 2013).

The effect of biochar is highly dependent on site and management conditions. In temperate soil-climate conditions, biochar has been shown to induce nutrient immobilization and, in some field trials, to slightly reduce crop yields (Brantley et al. 2015; Rodrigues et al. 2021). In conventional farming, nutrient immobilization can be compensated by the application of mineral N fertilizers, but this is not permitted in organic farming. Organic crop rotations are based on legume crops, biological N fixation and N transfer within the crop rotation (Pandey et al. 2017). If biochar application affects the growth and biological N fixation of grass-clover leys, this impacts the yield and N supply of the entire crop rotation (Brozyna et al. 2013).

The Roggenstein experimental field is characterized by predominantly fine-textured soil, a balanced pH, and high rainfall during the growing season, which collectively result in a yield potential that is exceptionally high (Mittermayer et al. 2024). In comparison with other organically managed sites, the yields from the unfertilized or compost-only treatments are high. However, when compared to the site's typical conventional yields (e.g., up to 11 t ha^{-1} of winter wheat grain), it is evident that the known yield ratio (organic/conventional) of approximately 0.6 for winter wheat is applicable as well (Döring and Neuhoff 2021).

The applied biochar C rates of 500, 1,000, and 2,000 kg ha^{-1} yr^{-1} are notably lower than in many other experiments where rates of up to 100 t ha^{-1} have been applied (Schmidt et al. 2021c). Schmidt et al. (2021b) propose much lower annual biochar application rates of 0.5–2 t ha^{-1} for agricultural practice. The longer the experiment lasts, the more the accumulated biochar will increase in the test plots. Numerous studies have demonstrated that biochar co-composting positively affects compost quality and reduces environmentally relevant emissions during the composting process (Ottani et al. 2022; Sanchez-Monedero et al. 2018). In many cases, this has reduced nutrient losses during composting, resulting in more nutrients being retained in the composting process when mass balance is considered (Agyarko-Mintah et al. 2017; Dias et al. 2010; Malinowski et al. 2019). However, this experiment examines the effects of compost use under realistic farm-level conditions, representing application rates typical for organic farms.

Biochar is incorporated into the compost piles at the outset of the composting process to ensure effective co-composting. Fertilization in the experiment is based on the amount of N applied, so the amounts of biochar applied differ slightly from the targeted application rates. During the long-term implementation of the field experiment, adjustments will be made to bring the actual quantities of C applied closer to the targeted values. The consistently high quality of organic fertilizers and reproducible experimental conditions are ensured by using constant feedstock materials and unchanged composting conditions.

Compost application increased legume and non-legume crop yields

Due to the low soil N availability resulting from decades of arable farming without application of organic fertilizers, the spelt yield was low in the first year of the experiment (Rapp et al. 2017). Despite the moderate compost N release in the year of application (Gutser et al. 2005), a significant yield increase was observed. Grass-clover compost had a slightly lower N use efficiency compared to manure compost. Due to the high N content of the compost feedstock, a N-rich compost was produced, which likely showed a higher N use efficiency than composts with lower N contents and a wider C:N ratio (Weber et al. 2014). The compost N release is also evident from the significantly

increased soil inorganic N stocks at the beginning of the growing season in all four experimental years (see Online Resource Tables S18-S21).

Grass-clover ley yield and N uptake reached a high site-specific level (Braun et al. 2010). Due to drought stress caused by the hot and dry summer, the third cut only achieved moderate yields (similar to the first cut). The small but cumulatively significant yield increase due to manure compost application can be explained by the supply of plant-available macro- und micronutrients (e.g. P, Cu, Fe, Mo), which are required for biological N fixation (Banerjee and Nath 2022). Furthermore, the higher proportion of weeds in the first cut and the higher proportion of grass in the second, third and fourth cut were attributed to the greater availability of plant-available N provided by the manure compost, which reduced the growth benefit of legumes (clover, alfalfa) over non-legume species and likely reduced the biological N fixation of the mixture (Harmoney et al. 2016; Kristensen et al. 2022).

Despite the good preceding crop value of grass-clover leys (Brozyna et al. 2013), a highly significant yield-increasing effect of compost fertilization on winter wheat was observed. The results demonstrate that, as in the initial phase of the experiment, N availability continues to constrain winter wheat grain yield, as evidenced by the fourth trial year with silage maize. The similar N use efficiency of manure and grass-clover compost amendment proves that the composts provide comparable levels of plant-available N regardless of the experimental year, thus confirming the initial hypothesis regarding the yield-increasing compost effect.

The use of NH_4-N-rich organic fertilizers, such as liquid manure or digestate, would likely have resulted in higher non-legume crop yields in this study for the same N input, as a larger proportion of the applied N becomes plant-available in the year of application (Gutser et al. 2005). However, over the long term, the two systems are likely to approach similar yield potentials due to an increased soil N mineralization potential from long-term compost use.

Biochar-related yield effects as a function of experimental duration

Both biochar applied alone and co-composted reduced grain yield and N uptake of spelt slightly but significantly, with more pronounced effects at larger biochar quantities applied. Other studies conducted in temperate climates with pristine biochar application also reported small to moderate yield reductions (Karer et al. 2013; Nelissen et al. 2014; Rodrigues et al. 2021), due to N immobilization or negative priming effects (Kloss et al. 2014; Weng et al. 2022). However, the effects of co-composted biochar were assessed contrarily in the literature. Some studies have found positive interaction effects resulting in higher crop yields due to improved nutrient availability from co-composted biochar (Gao et al. 2023; Kammann et al. 2015). Other studies have reported neutral or negative yield effects resulting from co-composting with biochar (Schmidt et al. 2014; Seehausen et al. 2017; Weldon et al. 2022). In addition to significant changes in the biochar surface characteristics, composting also results in alterations in the chemical properties of the compost and deposition of organic matter onto the biochar surfaces (Hagemann et al. 2017a; Khan et al. 2016). Furthermore, it has been demonstrated that biochar influences the composting process: In addition to increasing the composting temperature, it has been observed that the composting process proceeds more rapidly and is subject to fewer fluctuations due to unfavorable process conditions (Antonangelo et al. 2021). Moreover, humification was enhanced by biochar co-composting (Jindo et al. 2019; Sanchez-Monedero et al. 2018). An increase in composting temperature was also observed in this experiment (see Online Resource Table S6). It is conceivable that these processes result in the stabilization of non-pyrogenic compost organic matter (Casini et al. 2021) or physical entrapment of inorganic N within the porous biochar matrix (Hagemann et al. 2017b), and continuously inhibit compost N turnover after field application, resulting in minor yield reductions under the N-limiting conditions of organic agriculture.

Across all four cuts, pristine biochar had no effect, nor did co-composting result in significant effects on the yield and N uptake of the grass-clover ley. Farhangi-Abriz et al. (2022) elucidated the following mechanisms by which the application of biochar enhances the growth and biological N fixation performance of legumes: (1) improvement of chemical soil properties (cation exchange capacity, pH, soil organic carbon content), (2) enhancement of nutrient availability in the rhizosphere, and (3) creation of suitable

conditions for biological N fixation(including adequate soil moisture, soil pH, nutrient availability, temperatures (Ledgard and Steele 1992)). Considering the optimally adjusted pH of the fertile experimental soil for crop production, these mechanisms did not appear to be yield-limiting factors. However, the results did not indicate any adverse effects on crop yield. This aligns with the findings of Tammeorg et al. (2014), who demonstrated that legume growth under temperate soil-climate conditions is not significantly affected by biochar amendments. The findings indicate that the effects of biochar on legume growth are highly dependent on the study site and experimental conditions (Kätterer et al. 2019; Zhang et al. 2022). The botanical species composition of the grass-clover ley was not significantly influenced by the application of biochar, regardless of the co-composting process. Despite spelt demonstrating a minor but consistent yield-reducing biochar effect, a modest increase in legume proportions resulting from lower N availability may be plausible (Harmoney et al. 2016). This phenomenon did not occur, as N turnover was only slightly altered, with negligible effects on soil inorganic N stocks.

Despite the removal of the grass-clover ley aboveground biomass, substantial amounts of N-rich biomass and root residues remain in the field, resulting in considerable N mineralization after grass-clover ley incorporation (Davies et al. 2001). Despite its nutrient sorption properties (Nguyen et al. 2017), no yield-reducing effect of pristine biochar on winter wheat was observed (in contrast to spelt, 2021). Therefore, the N turnover of the grass-clover residues was presumably not significantly influenced. At present, there is a paucity of comparative data on the effect of biochar after grass-clover ley incorporation. Nevertheless, the yield-reducing effects of the co-composted biochar were only marginally observed in winter wheat. It is conceivable that the N supply from the compost amendments in previous years already outweighs the effects of reduced N use efficiency from the third compost application by co-composting with biochar (Gutser et al. 2005; Webb et al. 2013).

In contrast to the cereals grown in previous years, the co-composting with biochar consistently demonstrated superior maize biomass yields in the fourth year of the trial, depending on the quantity applied. The observed effect may be attributed to the accumulation of metabolizable N compounds remaining from previous trial years, now available to the maize plants (Romanyà et al. 2012). However, further investigation is required to ascertain whether this is due to crop-specific interactions between maize and biochar-compost blends, given that it has been demonstrated that maize roots enhance SOM mineralization (Kumar et al. 2016). Factors influencing soil N dynamics and N availability may be the longer cultivation period and the later N demand (June, July) compared to other cereal crops. In addition, plowing before maize planting, as was done in this study, often results in a significant increase in N mineralization (Winkhart et al. 2022). This means that maize benefits considerably more from soil N mineralization under nutrient-deficient growth conditions (Mittermayer et al. 2024). The forthcoming trial period will determine whether these positive yield effects can be achieved with other crops, as well as whether the initial yield reductions will have long-term beneficial implications.

It was initially presumed that co-composting with biochar would result in yield effects, and the biochar-only amendment reducing cereal grain yield. As these (significant) effects were observed only in the first trial year, these hypotheses are partially accepted.

Biochar and compost amendment increased SOC and TN stocks

After three years of experimental investigation, significant increases in SOC stocks were observed in all experimental treatments, including the unamended control. The cultivation of grass-clover ley is likely responsible for this observation, as it deposits substantial amounts of biomass C in the forms of crop and root residues, as well as their rhizodeposits, despite the removal of the aboveground biomass (Braun et al. 2010; Jensen et al. 2022). The application of organic fertilizers, including grass-clover and manure compost, resulted in a further increase in SOC stock, in addition to the effect found in the control treatment. This observation is consistent with numerous studies and meta-analyses demonstrating an increase in SOC stocks following compost and solid farmyard manure (compost) application (Maillard and Angers 2014; Poulton et al. 2018). Shortly, during the composting process of organic matter, humification processes occur, resulting in the enrichment of stable C

compounds (Chefetz et al. 1996; Leifeld et al. 2002). Nevertheless, precise data on the C accrual potential of grass-clover composts are currently not available. The pronounced SOC accumulation observed may also be attributable to the initially low SOC stock levels, providing a high potential for SOC accrual. In contrast, compost-related SOC accrual might be less pronounced in established organic farming systems with a long history of organic fertilizer application (Stewart et al. 2008).

Due to its aromatic structure, biochar remains nearly resistant to degradation in the soil over long periods (Smith 2016), depending on the pyrolysis conditions and the feedstock material (Singh et al. 2012). Therefore, it is not surprising that SOC stocks increased as a function of the applied biochar C rate, confirming the first hypothesis. It should be noted, however, that the C amounts applied closely match the C amounts found after soil analysis. This indicates that neither significant biochar decomposition processes nor losses due to leaching or wind drift occurred. The fact that some of the increases in SOC stocks are even higher than the amounts applied may be an indication of a negative priming effect and needs further trial duration and more sensitive analyses (Kuzyakov et al. 2000). This result might agree with the study by Blanco-Canqui et al. (2020). They observed an accrual of biochar-C, which was accompanied by an equivalent accumulation of labile C after six years of field experimentation.

When co-composted biochar was applied, the C fractions (compost and crop residue C, pyrogenic C) accumulated to considerable SOC stock increases (up to nearly 10 t ha^{-1}). This demonstrates that even under practice-relevant experimental conditions using relatively small biochar C application rates, substantial increases in SOC stocks can be achieved with repeated applications. This highlights the potential of biochar for SOC accrual (Don et al. 2024), which is hardly feasible with conventional methods (organic fertilization, cover cropping, multi-year forage cropping, reduced tillage (Beillouin et al. 2023). However, it must be considered that analyzing SOC stocks based on the fixed depth method may introduce biases due to soil compaction or expansion (Rovira et al. 2022). Nevertheless, the very similar bulk densities observed suggest that the soil mass studied remained largely unchanged.

The TN stock exhibited a slight decrease in the unfertilized control after three trial years. This phenomenon remains unexplained, as an increase in TN would have been expected following grass clover ley cultivation, which would be in line with the slight increase in SOC stock (Jensen et al. 2022). Compost amendment resulted in a significant increase in the TN stocks, compared to the biochar-only treatments. The crop N use efficiency of the composts used was approximately 10–30%, indicating that a substantial portion of the compost N added was transferred to the soil TN stock (Amlinger et al. 2003; Whalen et al. 2008). The accumulation of non-pyrogenic SOC is linked not only to C accrual, but also coupled with equivalent N input (Bhowmik et al. 2017). The notable increase in TN stocks resulting from biochar application (without composts) may be triggered by numerous mechanisms, including reduction in soil or crop residue N turnover, reduced N leaching, nitrate fixation or other processes that need to be further elucidated using more sophisticated approaches (Clough et al. 2013). The observed minimal decreases in crop N removal and N input from pristine biochar account for only a small fraction of the measured TN stock increase. A rise in TN stocks was also observed with increasing amounts of biochar in the composts. In addition to the mechanisms mentioned above, this is likely attributable to reduced inorganic compost N release. Nevertheless, it is important to note that the observed changes are still close to the detection limit.

Changes in SOC and TN stocks and their implications for future crop yields

The slight yield reduction observed was likely caused by a reduction in compost C and N turnover, which may have resulted in additional increase in SOC and TN stocks. These initial results may indicate the presence of negative priming effects (the accumulation of labile C and N compounds due to biochar-induced stabilization effects), consistent with the findings of Blanco-Canqui et al. (2020) and Weng et al. (2022). This interplay may result in enhanced non-pyrogenic SOC accrual in a longer term view (Janzen 2006) (in addition to biochar C accumulation) and could potentially reverse the observed biochar-induced yield reductions in a long-term perspective: Meta-analysis by Oldfield et al. (2019) has demonstrated that crop yields increase in conjunction with rising soil C

contents up to a certain level. In heterogeneous fields, crop yield has been shown to be closely related to SOC and TN stocks by increasing soil nutrient availability, N mineralization potential, microbial activity, and water-holding capacity (Mittermayer et al. 2022). Since the physical and chemical properties of pyrogenic (recalcitrant) and humic (labile) C are not comparable, however, it remains to be investigated whether there are interactions between these C fractions with implications for long-term yield effects. Therefore, the results indicate that enhancing crop yields in organic farming can only be increased by applying both components (compost + biochar, but not biochar-only), to enhance the soil N mineralization potential (Whalen et al. 2001).

Conclusions and outlook

The four-year field trial revealed time-dependent crop yield responses to biochar application. The application of co-composted biochar resulted in an initial slight but consistent reduction in crop yield, with positive yield effects, however, observed in the fourth year of experimentation. After three consecutive trial years, notable increases in SOC and TN stocks due to biochar and compost application were found. In organic arable farming, with N availability often limiting crop yields, the use of co-composted biochar in non-legume crops initially may result in slight yield reductions. However, compared to the notable yield effect of organic fertilizers, these yield reductions are small and are unlikely to be noticeable in agricultural practice due to the minor biochar quantities used. Nevertheless, it is conceivable that the slight yield reduction is confined to a few years following the initiation of biochar use in combination with compost. Yet, there may be potential to boost long-term crop yields via SOC accumulation from biochar C and labile C inputs, as well as through a gradual increase in soil N mineralization potential resulting from initially reduced compost N release. It can be concluded that determining the optimal mixing ratios of biochar and compost will require a longer trial period to achieve reliable results. This study highlights the importance of multi-year field experiments in analyzing SOC and TN dynamics and understanding their effects on crop yield. Such biochar-related studies are currently lacking, yet they are indispensable; no laboratory experiment or model simulation can fully replicate these complex interactions. Therefore, this experiment will be continued, anticipating deeper insights into SOC and TN accumulation processes and long-term yield dynamics. Subsequent studies will therefore focus on changes in soil properties, such as potential interactions between labile and pyrogenic C pools, while also assessing the impact on long-term crop yield.

Acknowledgements The authors would like to express their gratitude to Stefan Kimmelmann, Jan Dirk Otten, Johann Ludwig (Experimental Station Roggenstein, TUM Plant Technology Center), and all others involved in the accurate and professional conduct of this complex field trial. Additionally, we would like to acknowledge Josef Reif for providing the high-quality manure and the Greenhouses & Phytochambers Unit (TUM Plant Technology Center) for the use of the compost area. We would like to express our gratitude to Marcel Huber and Benjamin Hupfauf (SynCraft Engineering GmbH, SIOTUU GmbH) for providing the biochar and for their helpful input. We would also like to thank Tabea Reisländer, Lukas David, Johanna Merkle, and all other student assistants for their invaluable assistance with data collection in the field trial and during composting. We would also like to thank Agathe Zach, Monika Kaindl, Dr. Ludwig Nätscher and Armin Stich for their conscientious laboratory analysis of the soil and plant samples. Finally, we would like to express our gratitude to Stephan Haug (TUM|Stat) for his valuable insights in the statistical analysis of the experimental data and to Ernst-Pelz-Stiftung e.V. (Geretsried, Germany) for their generous financial support during the first year of the trial.

Author contributions Emanuel Jaufmann: Conceptualization, Methodology, Investigation, Data Curation, and analysis, Writing—original draft. Harald Schmid: Conceptualization, Methodology, Writing —review & editing. Kurt-Jürgen Hülsbergen: Conceptualization, Methodology, Supervision, Writing—review & editing. All authors read and approved the final manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL. This study was funded by the Bavarian State Ministry of Food, Agriculture, Forestry and Tourism, (Bayerisches Staatsministerium für Ernährung, Landwirtschaft, Forsten und Tourismus), Research project “TerraBayt”.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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